

Supplementary Methods: object-switch trajectory analysis and human-AI workflow

Heng Zhao, Yufei Wang, Ruidian Song and Weisheng Chen

1. Empirical-object hierarchy

The study separates three levels that are easily conflated.

1. **Implementation objects:** model weights, attention heads, MLPs, residual blocks and layers. These are the engineered objects that realize inference.
2. **Empirical process object:** the ordered hidden-state sequence $T_{m,p} = (h_{m,p,1}, \dots, h_{m,p,L_m})$. This is the object on which the present mechanism comparisons are defined.
3. **Coordinates and diagnostics:** direct hidden summaries, hidden-state flow, vocabulary-facing TopK/VIM readbacks, the decision-window clean-relative margin profile, ΔU , transition coordinates and residual diagnostics.

The methodological statement "empirical research object before coordinates" means choosing a testable process object before selecting measurements. It does not claim that engineering objects are unimportant or make a metaphysical claim about the layer-state trajectory.

H-shape denotes the ordered profile of clean-relative candidate margins across a specified layer window. It is a trajectory-shape summary, not a hidden-state tensor or an independent mechanism label. Earlier analyses also used related shorthand for window summaries and previous-transition history; those objects are kept distinct here. The decision-window H-shape is exactly dR over decision layers and is construction-aligned with ΔU . Legacy labels remain only in the Source Data manifest for provenance.

2. Intellectual provenance and operational boundaries

Manuscript object	Mathematical or physical source idea	Operational use here	Identity explicitly not claimed
Layer-state trajectory	State trajectory of a discrete dynamical system	Ordered sequence of last-token hidden states across blocks	Layer depth is not physical time
Exploratory registration	Functional or phase registration	Data-selected windows provide model-specific indexing choices for downstream analyses	Prospective joint-role recurrence failed; windows are not confirmed phases or identical coordinates
ΔU	Reaction coordinate and order-parameter compression	Oriented PC1 of a clean-relative decision-window margin matrix	Not energy, potential, thermodynamic order parameter or full mechanism
Transition closure	State-space and reduced-order closure	Test whether current-state and realized-transition coordinates compactly describe the observed update	Not an autonomous dynamical law when target-transition information enters
Chord-directed geometry	Finite-path alignment, detour and turning	Compare real order with endpoint-preserving shuffles under a declared cosine geometry	Not a learned metric, exact geodesic, least action or conservation law
Structured residual	Residual diagnostics in reduced-order modelling	Test whether deviation from the chord-directed reference retains information	Not a complete tangent-normal decomposition
Local-transition actuator	Feedback-control distinction between state estimation, abstention and actuator choice	Mechanism-conditioned transition operator plus confidence-gated bounded action	Crosses a declared emitted-token boundary in three checkpoints; not answer-correctness, safety or deployment control

The source ideas were introduced after empirical problems required a compact description. They were not used as labels that made an observation true. Negative results narrowed each analogy.

3. Controlled prompt subset

The direct hidden-state analyses used eight complete graph groups selected with seed 20260610: group IDs 0, 2, 15, 23, 25, 26, 51 and 63. All 12 conditions per group were retained, giving 96 prompts. The same source rows were used for all three checkpoints. This design preserves within-group condition structure while making 50 layer-order shuffles tractable. It is not a first-row convenience sample.

The subset manifest contains prompt ID, group and graph IDs, raw condition, analysis label, task-defined clean and conflict candidates, source row and model-specific readback records. Final generated-answer correctness is not used in ΔU , coordinate-audit labels, intervention-class labels or policy labels.

Each prompt encodes a short relation chain from named entities to one of two declared answer candidates. The clean candidate is supported by the unperturbed chain. The conflict candidate is introduced by a distractor, competing statement, update or exception. Stable conditions retain a decisive clean chain, competition conditions support both candidates, and closure conditions redirect the chain. These labels are fixed by prompt construction before model fitting. They are analysis classes, not dynamical states discovered from the hidden trajectories. The task was selected to isolate evidence competition and updating under controlled counterfactual conditions; no claim is made that it represents open-ended reasoning.

Supplementary Methods Table 1 | Analysis-label and information-boundary crosswalk.

Analysis family	Source object and raw conditions	Fitted target or endpoint	Inputs permitted	Information explicitly excluded
Coordinate audit and 96-prompt geometry subset	Twelve-condition panel. Stable: clean, rename, irrelevant, paraphrase and weak distractor (stored as stable-shift). Competition: balanced competition, direct conflict, equal evidence and source claim (stored as source-ambiguity). Closure: closure update, closure override and exception override.	Three-class prompt-condition label for proxy comparison or plot grouping	Prompt metadata for labels; fold-fitted predecision state and the audited proxy for prediction	ΔU , decision-window clean-relative profile, generated answer and correctness
Functional-window scan	Separate 12-condition discovery panel. Stable: clean, rename, permuted, redundant, irrelevant and paraphrase; weak distractor stored as stable-shift and collapsed with stable. Competition: balanced competition and direct conflict. Closure: closure update, closure override and exception override.	Exploratory preservation, class-information and downstream-coupling score	Candidate window measurements within graph-aware folds	Confirmatory interpretation; selection is not nested
Prospective functional-window boundary	Same 12 templates on 24 new graph groups with disjoint entities, relation phrases and candidate labels	Frozen-window role recurrence and rank	Discovery-only PCA and fitted models; confirmation labels only for final scoring	Window reselection or confirmatory claims after a failed gate
ΔU construction	Declared clean and conflict candidate identities within each graph group	Oriented PC1 score of the clean-relative decision-window matrix	Candidate margins in the declared window; training-fold PCA where transfer is evaluated	Generated answer, final correctness and an analysis-independent mechanism label
Internal-control and confidence-gated intervention	Separate ten-condition panel. Stable: clean, redundant, irrelevant and paraphrase. Competition: weak distractor, balanced competition and direct conflict. Closure: closure update, closure override and exception override.	Three-class intervention label used by the mechanism classifier and gate	Training-graph prompt labels and pre-intervention trajectory features	Held-out outcomes, generated answer and correctness
Policy supervision	Eight-action internal library or expanded ten-action boundary library	Discrete (α, β) action maximizing the declared training utility; none is $(0, 0)$	Training-graph intervention responses and fitted mechanism features	Held-out action outcomes and correctness
Candidate-boundary endpoint	Paired no-intervention and controlled continuations	Conflict-to-clean crossing of the two declared candidates; in the expanded test, strict full-vocabulary top-1 crossing	Held-out outputs only for evaluation	Use as a fitting label; interpretation as correctness or open-ended answer control

The coordinate-audit and intervention panels therefore do not share a universal latent class variable. In particular, weak distractor is stored as stable-shift and collapsed with stable in the coordinate audit, but is grouped with competition in the separately generated intervention panel. Both mappings were fixed in their corresponding data-generation and analysis scripts before model fitting. They are retained rather than harmonized after observing outcomes.

4. Direct hidden-state update summaries for Fig. 1

For normalized states $x_l = h_l / \|h_l\|_2$, adjacent hidden-state change is

$$c_l = 1 - \cos(x_l, x_{l+1}).$$

Let $s_l = x_{l+1} - x_l$ and hidden size d . The normalized participation ratio is

$$P_l = \frac{(\sum_{j=1}^d s_{l,j}^2)^2}{d \sum_{j=1}^d s_{l,j}^4}.$$

P_l approaches one when step energy is evenly distributed and decreases when fewer dimensions dominate. Fig. 1 reports means and standard errors by collapsed coordinate-audit condition family and checkpoint. These families are predefined plot groups, not inferred trajectory phases. The calculations use native hidden states and no vocabulary projection. Executable filenames are listed in the Source Data manifest.

5. Readback and projection-selection controls

For clean candidate token c_p , conflict candidate token e_p and output matrix W ,

$$R_{p,l} = h_{p,l}^\top (W_{c_p} - W_{e_p}).$$

TopK/VIM features summarize projected vocabulary neighbourhoods. The row-permuted control shuffles rows of W , preserving matrix values but changing token correspondence. The Gaussian control uses a seeded random matrix of matched shape. Because these controls operate after hidden-state extraction, they audit coordinate dependence of the vocabulary-facing geometry without changing the native hidden-state trajectories.

For the independent prompt-class audit, every proxy augmented the same predecision base state. Proxy PCA was capped at 32 dimensions and fitted separately inside each training fold. Five-fold GroupKFold used relation graph as the split unit. In the 12-condition source panel, clean, rename, irrelevant and paraphrase were stored as stable; weak distractor as stable-shift; balanced competition, direct conflict and equal evidence as competition; source claim as source-ambiguity; and the three update or override conditions as closure. Stable and stable-shift then mapped to the stable classifier class; competition and source-ambiguity mapped to competition; closure mapped to closure. Macro F1 is the unweighted mean of class-specific F1 scores. The reported value is augmented minus base prompt-class macro F1. The label map is fixed by controlled prompt metadata and does not use ΔU , generated answers or the decision-window clean-relative margin profile.

Algorithm 1: independent coordinate audit

for each checkpoint and proxy family:

 partition relation graphs into five folds

 for each held-out fold:

 fit all preprocessing, including proxy PCA, on training graphs only

 fit the base mechanism classifier on predecision state features

 fit the augmented classifier on the same base plus the proxy

 predict the held-out graphs

 concatenate held-out predictions across folds

 report macro-F1(augmented) - macro-F1(base)

6. Exploratory functional-window scan and prospective boundary test

Candidate windows span the first 45% of checkpoint depth. Window lengths are 2, 3, 4, 5, 6 and 8, step size is one layer, and TopK neighbourhood sizes are 100 and 500. For each window, the preservation score is one minus the mean clean-relative distance for relation-preserving conditions. The source field was named topology, but the quantity does not compare relation-preserving with structure-changing conditions and is not itself a topology-separation test. Prompt-condition-class information is GroupKFold macro F1, and downstream coupling is a GroupKFold correlation with ΔU . The objective is

$$S = 0.5 S_{\text{preservation}} + 0.8 \max(S_{\text{class}}, 0) + 0.8 \max(S_{\text{downstream}}, 0).$$

The maximum-scoring interval is retained. Selection is non-nested because the same scan supplies candidate and selection score. GroupKFold prevents graph overlap in component estimates but does not remove selection optimism. The windows are exploratory and data-selected.

Algorithm 2: exploratory functional registration

for each checkpoint:

```

enumerate all permitted starts, lengths and neighbourhood sizes

for each candidate window:

    calculate clean-relative preservation

    calculate group-held-out prompt-condition-class macro F1

    calculate group-held-out downstream correlation

    combine the three quantities using the stated score S

retain the maximum-scoring window

label the result exploratory because selection and evaluation share the scan

```

6.1 Prospective confirmation protocol and result

After the exploratory scan, the overall windows and neighbourhood sizes were frozen as Qwen layers 5-6 with $K = 500$, Llama layers 0-2 with $K = 500$, and Gemma layers 1-6 with $K = 100$. The prospective confirmation set comprised 24 new relation graphs and the same 12 perturbation templates. It replaced every discovery entity triple and relation pair and used 12 candidate labels drawn from an audited common one-token vocabulary that was disjoint from the discovery candidate labels. No confirmation prompt entered the original scan.

For each checkpoint, the discovery rows alone fitted the clean-relative decision-profile PCA, its sign orientation, feature scaling, a balanced three-class logistic classifier and ridge regression to ΔU . The fitted objects were then applied to confirmation rows. Confirmation labels were used only to score the predeclared endpoints. The full candidate pool was evaluated to obtain the rank of the already frozen window; no new winner replaced it. Two thousand graph-group bootstraps produced 95% intervals. Nine hundred and ninety-nine discovery-label permutations formed the class-information null, and 999 within-discovery-graph ΔU permutations formed the downstream null.

The frozen per-checkpoint gate required: (i) a positive 95% bootstrap interval for the graph-paired difference between structure-changing and relation-preserving composite distances; (ii) macro F1 at least 0.55 and greater than the label-null 95th percentile; and (iii) positive downstream correlation greater than the target-null 95th percentile. Window-location recurrence was declared separately when the frozen window ranked in the top quartile of the unchanged candidate pool. The protocol and its SHA-256 hash were written before extraction. One initial run stopped after Qwen extraction because the imported feature function returned two objects rather than three; no endpoint was calculated. The interface correction and failed-run logs are retained.

Supplementary Methods Table 2 | Prospective functional-window confirmation on 24 untouched graph groups.

Checkpoint	Topology separation (95% bootstrap interval)	Class macro F1 / label-null 95th	Downstream r / target-null 95th	Frozen rank	Joint role gate	Location gate
Qwen	-0.185 (-0.189, -0.181)	0.739 / 0.521	0.659 / 0.623	58/112	Fail	Fail
Llama	-0.092 (-0.092, -0.091)	0.662 / 0.510	0.411 / 0.452	38/40	Fail	Fail
Gemma	-0.090 (-0.091, -0.089)	0.649 / 0.475	0.624 / 0.404	5/100	Fail	Pass

The class-information endpoint transferred in all checkpoints. Downstream coupling passed its frozen null in Qwen and Gemma but not Llama. Topology separation reversed in every checkpoint, and only Gemma retained top-quartile location. The three-checkpoint role-recurrence claim is therefore rejected. These results do not invalidate use of the discovery windows as explicitly exploratory analysis choices, but they prohibit treating the windows as confirmed functional phases or homologous cross-checkpoint locations.

7. Executable ΔU definition

Within each graph group, the clean condition supplies a baseline. If a clean row is absent, the group mean is used. The clean-relative margin is

$$dR_{p,l} = R_{p,l} - R_{p_0,l}.$$

For mapped decision layers $\mathcal{L}_{\text{dec}}$, form

$$D = [dR_{p,l}]_{p,l \in \mathcal{L}_{\text{dec}}}.$$

PCA is fitted to D , and ΔU_p is its PC1 score. The sign is oriented so that stable or stable-shift rows have greater mean scores than closure rows when those groups are available. Candidate identities from the controlled task enter R ; the model's generated answer and final correctness do not. ΔU is therefore a task-constructed exploratory diagnostic of declared candidate-path separation. Its numerical axis is not transferred to a different prompt family. The controlled lexical and addition tests fit new task-specific coordinates with the same protocol and report held-out separation independently.

The decision-window clean-relative margin profile is $D_{p,\cdot}$. It cannot serve as independent evidence for ΔU because both are the same construction at different dimensionalities. Correlation with a dominance readout derived from related decision-window information is a sanity check. Independent bounding analyses remove the direct margin, exclude decision and endpoint information, change the window, test residual signal or compare intervention nulls.

8. Transition-coordinate closure

Previous-transition history features comprise the preceding clean-relative margin change, preceding rank-gap change, reversals in both signals, and backtracking relative to the preceding and initial centres. The realized transition coordinate O_l^{cont} comprises the first ten principal components of the observed transition representation. Closure models compare state-only, history-only, combined state-history, state plus O_l^{cont} , and bilinear state-transition feature sets.

If adjacent target-layer information enters O_l^{cont} , the analysis is descriptive factorization. Macro R^2 is the unweighted mean of target-wise R^2 values. The predictive audit estimates $\hat{O}_l^{\text{cont}}$ from current-state, previous-transition history and optional layer/condition context under graph-grouped cross-validation, excluding the realized target transition. Within each outer fold, the first-stage model predicts $\hat{O}_l^{\text{cont}}$ separately for training and test rows. The second-stage model then constructs state-coordinate products independently from each fold's own predictions; no observed O_l^{cont} enters that stage. State-only closure reached macro $R^2 = 0.822$, observed O_l^{cont} reached 0.979, and the best target-excluded bilinear model using $\hat{O}_l^{\text{cont}}$ reached 0.862. Its increment over state only was 0.0394 (95% prompt-group bootstrap interval, 0.0368-0.0426), and it retained 25.1% of the observed-coordinate gain. Models predicting O_l^{cont} itself ranged from macro $R^2 = 0.41$ to 0.61. All four corrected bilinear feature sets completed five graph-grouped folds and improved over their matched additive versions by 0.0279-0.0318 macro R^2 . This is target-excluded incremental prediction, but remains exploratory and does not establish a task-general forward law. Metric-specific embedding diagnostics are not combined because distance and correlation scores have different scales and interpretations.

Prospective final candidate-margin forecast

This separate analysis predicts the exact final clean-minus-conflict output-logit margin from partial decision-window histories. The frozen relation-graph split contains 67 training groups (670 prompts) and 29 held-out groups (290 prompts). At cutoff layer k , the ordered-history predictor contains only the candidate-margin values observed at decision layers $l \leq k$. It excludes all later states and transitions, the final generated token and answer-correctness labels. Ridge regression uses penalty 10. A balanced logistic regression with $C = 1$ predicts whether the final candidate margin is positive from the same features.

For the order null, the current-cutoff value is fixed and all earlier values are independently permuted within each prompt. This avoids the feature-renaming invariance of a common permutation followed by model refitting. Fifty null datasets are independently refitted and evaluated on the unchanged held-out graph groups. Confidence intervals use 1,000 relation-graph bootstrap replicates. The earliest real-history cutoff above the null R^2 95th percentile is Qwen layer 24 ($R^2 = 0.574$, AUROC = 0.857), Llama layer 14 ($R^2 = 0.393$, AUROC = 0.837) and Gemma layer 21 ($R^2 = 0.714$, AUROC = 0.984). This is a bounded prospective forecast of a declared candidate contrast. It is not an autonomous state equation or a general model of answer correctness.

9. Direct hidden-state chord geometry

For normalized hidden states, the endpoint chord is $g = x_L - x_1$. Chord alignment is the mean cosine between each step s_l and g . Path length is the sum of adjacent cosine distances, endpoint distance is the first-to-last cosine distance, detour is their ratio, and turning curvature is the mean angle between adjacent steps.

Each null retains x_1 , x_L and the complete intermediate state set but randomly permutes intermediate layer order. One permutation is shared across all 96 prompts within a checkpoint, and 50 permutations are evaluated. The 50 checkpoint-level means form the null distribution; prompt-level p values are not pooled. Alignment gain is real minus shuffle mean. Detour and curvature reductions are $\log_2(\bar{q}_{\text{shuffle}}/\bar{q}_{\text{real}})$. Positive reductions favour observed order.

For a metric where larger values favour observed order, the null-standardized effect is

$$Z_q = \frac{q_{\text{real}} - \text{mean}(q_{\text{null}})}{\text{sd}(q_{\text{null}})}.$$

For detour and curvature the numerator is reversed so that positive values again favour observed order. The plus-one empirical one-sided p value is $(1 + n_{\text{extreme}})/(50 + 1)$. All three observed checkpoint means were beyond all 50 corresponding alignment null means, giving the minimum attainable empirical value of 1/51; this finite resolution is reported rather than interpreted as a more precise p value. The standardized separation was 5.11, 5.57 and 5.63 null standard deviations for alignment; 11.87, 6.31 and 11.50 for detour; and 6.03, 4.20 and 6.42 for curvature in Qwen, Llama and Gemma, respectively.

Algorithm 3: endpoint-preserving layer-order null

for each checkpoint:

 compute one real checkpoint mean from the 96-prompt controlled subset

 repeat 50 times:

```

draw one permutation of intermediate layer indices

apply that same permutation to every prompt

retain the first and final layer states

compute the checkpoint mean for each geometry metric

compare the real mean with the 50 repeat-level means

```

The estimand is limited to normalized hidden states, cosine geometry and this endpoint-preserving null. It does not test an exact geodesic equation. A fixed-seed rerun reproduced the archived output hashes; the hashes and filenames are recorded in the Source Data manifest.

10. Independent task-family and random-initialization controls

10.1 Controlled lexical and addition tasks

The independent task family asks which of two natural words belongs to a declared semantic category. It contains 64 problem groups and seven category inventories: colours, directions, metals, shapes, landscape features, sky objects and fruits. Candidate words were drawn from a 28-word intersection that tokenized as one continuation token in Qwen, Llama and Gemma. Correct-candidate display position alternated by problem identifier. Two stable prompts, two prompts containing a competing annotation and two correction prompts were generated for each problem, giving 384 prompts per checkpoint. The prompts used each checkpoint’s native chat template.

The external task inherited the frozen relation-graph decision windows: layers 20-25 in Qwen, 12-15 in Llama and 18-23 in Gemma. No layer scan or task-specific window selection was performed. Problem groups were divided once with seed 20260714 into 44 training and 20 held-out groups. For each problem, the clean prompt supplied the candidate-margin baseline. PC1 was fitted to the training-group clean-relative decision profiles and applied to the held-out groups. A balanced multinomial logistic classifier used only this one-dimensional coordinate. Its held-out macro F1 was compared with 200 fits in which training labels were permuted while held-out labels and the split were unchanged.

The direct hidden-state geometry analysis retained all 384 prompts and used 50 endpoint-preserving intermediate-layer permutations per checkpoint. Real-order alignment exceeded all 50 null means in each checkpoint. Held-out lexical-coordinate macro F1 exceeded the label-null 95th percentile in Qwen and Llama but not Gemma. Clean-condition candidate selection was 0.891, 0.547 and 0.953, respectively, so the Llama result is retained with a weak-behaviour boundary.

The controlled addition task contains 64 single-step addition problems and the same six-condition design. It uses the same frozen decision windows, 44/20 problem split, train-only PC1 construction, 200 label shuffles and 50 endpoint-preserving order shuffles. Geometry exceeded all order nulls in all checkpoints. Held-out coordinate macro F1 was 0.583, 0.456 and 0.570 in Qwen, Llama and Gemma, above the corresponding label-null 95th percentiles. Llama pair accuracy was 0.315, so the coordinate result does not imply strong arithmetic behaviour. An earlier mixed-operation version is retained as a coverage audit rather than a primary result.

10.2 Standard four-choice ARC control

The standard benchmark control used the allenai/ai2_arc ARC-Challenge validation split. The 96-item audit subset comprised the earliest validation rows in dataset order whose answer keys were exactly A-D. This deterministic eligibility rule was fixed without reference to checkpoint outputs, bounding the extraction budget while preventing outcome-based item selection. Every original question and all four answer choices were presented through the checkpoint-native chat template, and the model was instructed to return one option letter. No binary clean/conflict wrapper was introduced. Four-choice accuracy was 0.781, 0.333 and 0.771 in Qwen, Llama and Gemma.

Hidden states were extracted using the same last-token rule as the main analysis. For each checkpoint, 50 endpoint-preserving intermediate-layer shuffles supplied the null. Real-order alignment exceeded all 50 null means, with standardized separations of 3.21, 5.23 and 7.84. The deterministic subset is an audit set rather than a new benchmark estimate. This analysis tests the order geometry beyond binary prompt structure; it does not transfer ΔU , mechanism labels or the intervention policy.

10.3 Random-initialization control

The random control used the exact Qwen2.5-1.5B, Llama-3.2-1B and Gemma-2-2B model configurations and native tokenizers. Three models per configuration were instantiated with seeds 1701, 1702 and 1703; token embeddings, attention and MLP parameters, normalization parameters and output heads were random, and no trained tensor was loaded. The inputs were the same fixed 96 raw relation-graph prompts used in the trained geometry audit. Hidden states were extracted after every transformer block using the same last-token rule.

Each parameter state was compared with 50 endpoint-preserving layer-order permutations using the same metric definitions and null seed 20260717. Mean random alignment gains were 0.0768, 0.1028 and 0.0879 in Qwen, Llama and Gemma, versus 0.0317, 0.0403 and 0.0499 in the trained checkpoints. Fig. 4 used a separately seeded 50-shuffle null; its small gain differences arise from the sampled null mean, while the trained real-order alignment is identical. All nine random gains were positive and exceeded their matched trained checkpoint. The result establishes an architecture-compatible order effect across these configurations. It does not show that trained and random trajectories carry the same function or identify a universal architecture-training decomposition.

11. Auxiliary readback-centre geometry

This auxiliary analysis operates on normalized TopK/VIM centres C_l , not raw hidden states. It uses the same broad family of endpoint-chord, cosine-path, detour and turning diagnostics and fixes endpoints while shuffling intermediate readback-centre layers. It is retained as a cross-observable diagnostic, and its estimand is never labelled hidden-state geometry.

12. Intervention labels, features and splits

Intervention-class labels are fixed by prompt condition in the separately generated ten-condition control panel. Stable contains clean, redundant, irrelevant and paraphrase; competition contains weak-distractor, balanced-competition and direct-conflict; closure contains closure-update, closure-override and exception-override. These are controlled intervention families rather than correctness labels or trajectory-derived dynamical phases. Their mapping is not substituted into the 12-condition coordinate audit.

Mechanism features are the predicted candidate-path separation and its magnitude, a companion dominance estimate, velocity magnitude and standardized velocity, normalized layer depth, a precursor-layer indicator, and baseline separation and dominance values. Policy features add mechanism probabilities and predicted-mechanism indicators. Training and test partitions are separated by relation graph, preventing rows from one graph from appearing in both partitions. Final correctness is not a target.

The first two intervention families used additive state directions, and the third used an additive flow direction. Their failure to yield robust mechanism-specific selectivity motivated the switch to a local mechanism-conditioned transition operator. At each intervention layer, principal-component analysis was fitted to the concatenated current and next hidden states from training graphs. Its dimension was $r_l = \min(128, 2n_{\text{train}} - 1, d_m)$. In the common PCA coordinates, separate stable and closure maps solved

$$(A_l^{(k)}, b_l^{(k)}) = \underset{A, b}{\operatorname{argmin}} \sum_{i \in \mathcal{T}_k} \|z_{i, l+1} - Az_{i, l} - b\|_2^2 + 10 \|A\|_F^2,$$

for $k \in \{\text{stable}, \text{closure}\}$. The intercept was not penalized. The stable-row prediction was inverse-transformed to hidden-state space to define $\hat{F}_l^{\text{stable}}$. For realized next-layer state $h_{p,l+1}$, the intervention is

$$\tilde{h}_{p,l+1} = (1 - \alpha_{p,l})h_{p,l+1} + \alpha_{p,l}\hat{F}_l^{\text{stable}}(h_{p,l}),$$

$$h_{p,l+1}^{\text{int}} = \tilde{h}_{p,l+1} + \beta_{p,l} \text{sd}(\tilde{h}_{p,l+1})d_l,$$

where d_l is the unit training-graph ridge direction for ΔU . The operator interpolation is applied first. The action grid is

$$\mathcal{A}_0 = \{(0, 0), (0.1, 0.1), (0.2, 0.2), (0.3, 0.3), (0.3, 0), (0, 0.3), (0.3, 0.2), (0.2, 0.3)\},$$

and **none** denotes $(0, 0)$. For training prompts, closure labels maximize signed ΔU shift; stable and competition labels minimize absolute shift. The final policy is compared with matched fixed combinations and 50 mechanism-label, policy-label and joint shuffles.

The mechanism classifier used the nine-dimensional vector

$$x^{\text{mech}} = (\widehat{\Delta U}, |\widehat{\Delta U}|, \widehat{D}_B, \|v\|, z_v, \tilde{l}, I_{\text{precursor}}, \Delta U_{\text{base}}, D_{B,\text{base}}).$$

It was a 250-tree `RandomForestClassifier` with $\text{max_depth} = 7$, $\text{min_samples_leaf} = 6$, $\text{class_weight} = \text{balanced}$, $\text{subsample} = 0.5$ and seed $42 + s$, where $s = 0$ for the real fit and the shuffle index otherwise. The policy input appended the three predicted mechanism probabilities and three one-hot predicted-mechanism indicators, giving $x^{\text{policy}} \in \mathbb{R}^{15}$. The policy was a 200-tree random forest with $\text{max_depth} = 6$, $\text{min_samples_leaf} = 8$, the same class weighting and seed $142 + s$. Both used the scikit-learn default weighted-Gini split criterion. They contained no neural hidden layers and used neither gradient descent nor policy gradients.

For every training prompt, each action $a = (\alpha, \beta)$ in the eight-element grid was executed. The discrete target was

$$a_p^* = \begin{cases} \arg\max_a [\Delta U_p^{(a)} - \Delta U_{p,\text{base}}], & p \in S_{\text{closure}}, \\ \arg\min_a |\Delta U_p^{(a)} - \Delta U_{p,\text{base}}|, & p \notin S_{\text{closure}}. \end{cases}$$

The policy forest learned this discrete label on layer-expanded training rows and predicted one action per held-out prompt-layer row. Thus the policy is supervised action classification after an exhaustive training-grid evaluation, not reinforcement learning.

The mechanism-label null permutes training mechanism labels, refits the mechanism classifier and then refits the downstream policy. The policy-label null retains the real mechanism classifier but permutes training action labels before policy fitting. The joint null independently permutes both label sets and refits both classifiers. All nulls retain the graph split, test rows, transition operators, action grid and endpoint calculation.

Executable procedure: held-out mechanism-conditioned intervention

split complete relation graphs 70:30 into training and held-out sets

fit transition operators, mechanism classifier and action policy on training graphs

for each held-out prompt and selected layer:

predict mechanism probabilities

select operator and precursor weights

apply operator interpolation, then the scaled precursor update

measure post-intervention Delta U and the declared candidate logits

compare the learned policy with matched fixed actions and refitted label-shuffle nulls

Specificity is

$$s(m, c) = \text{mean}_{i \in S_{\text{closure}}} |\Delta U_i^{(c)} - \Delta U_{i,\text{base}}| - \text{mean}_{i \in S_{\text{nonclosure}}} |\Delta U_i^{(c)} - \Delta U_{i,\text{base}}|.$$

The pseudo-log transform with $\sigma = 0.08$ is used only for Fig. 5 display. The primary endpoint of this original action-budget analysis is internal dominance specificity. Because the intervention is applied upstream of this held-out endpoint and is compared with fixed and refitted label-shuffle controls, it supports selective causal control of the measured internal answer-formation trajectory. Output-boundary control is tested separately below with an expanded, training-only action search and a strict full-vocabulary endpoint.

The candidate-choice boundary pairs no-intervention and real-policy outputs for each held-out prompt. It reports the change in final clean-minus-conflict candidate margin, the change in binary choice between those two candidates and their association with ΔU shift. Intervals for these paired descriptive contrasts use 5,000 relation-graph bootstrap replicates with seed 20260714. In closure prompts, clean denotes the original unperturbed-chain candidate, whereas the updated or override-supported conflict candidate is generally the rule-consistent answer. A conflict-to-clean crossing is therefore an actuator-response endpoint, not an accuracy improvement.

Out-of-fold margin-shift and boundary-crossing analysis

The quantitative bridge defines

$$\delta M_p = M_{p,\text{policy}} - M_{p,0}, \quad \delta \Delta U_p = \Delta U_{p,\text{policy}} - \Delta U_{p,0}.$$

For checkpoint m and regime r , an affine ridge model with penalty 10^{-6} estimates

$$\widehat{\delta M}_p = a_{m,r} + b_{m,r} \delta \Delta U_p.$$

Five-fold GroupKFold holds out complete relation graphs, with the same graph held out across checkpoints. The predicted post-policy margin is $\widehat{M}_{p,\text{policy}} = M_{p,0} + \delta \widehat{M}_p$. For the at-risk population $M_{p,0} < 0$, the observed crossing label is $I(M_{p,\text{policy}} \geq 0)$. Within each outer fold, an inner graph cross-fit supplies training predictions for a one-variable logistic calibrator, which maps predicted post-margin to crossing probability. Thus neither the margin-shift prediction nor its probability calibration evaluates a graph used for fitting.

The pooled closure margin-shift model reached $R^2 = 0.850$ with a 95% graph-bootstrap interval of 0.811-0.882. Among 188 at-risk closure prompts, 15 crossed the candidate boundary and pooled AUROC was 0.976 (0.951-0.996). Under the original action budget, the observed counts were 0/87 in Qwen, 9/45 in Llama and 6/56 in Gemma. Qwen’s mean shift was largest in absolute logit units, but its mean initial deficit was 5.03; 94.0% of its closure-layer actions used the maximum (0.3, 0.3) action. Continuous and crossing intervals use 1,000 graph-bootstrap replicates with seed 20260714. Internal trajectory control was observed in all three checkpoints, whereas candidate-boundary crossing under the original action budget was checkpoint-dependent. This endpoint is not open-ended generated-answer correctness.

Confidence-gated emitted-token boundary control

The expanded output-boundary analysis was run independently in Qwen, Llama and Gemma with the same seed-42 split of complete relation graphs: 67 training graphs containing 670 prompts and 29 held-out graphs containing 290 prompts, of which 87 were closure prompts. Intervention layers were 15-19 in Qwen, 8-11 in Llama and 13-17 in Gemma. The principal-component direction defining ΔU , transition operators, precursor directions, mechanism classifier, action targets and boundary policy were fitted on training graphs only. Candidate token identities entered the paired-margin objective. Generated answer text, final-answer correctness and held-out candidate outcomes entered neither fitting nor gate construction.

The bounded action library was

$$\mathcal{A} = \{(0, 0), (0.3, 0.3), (0.6, 0.3), (0.6, 0.6), (0.9, 0.3), (0.9, 0.6), (0.9, 0.9), (1.0, 0.6), (1.0, 0.9), (1.0, 1.2)\}.$$

For action $a = (\alpha, \beta)$, the state update retained the operator-then-precursor order defined above. Because $0 \leq \alpha \leq 1$, operator interpolation could reach but not extrapolate beyond the fitted transition proposal. On a training closure prompt p , action utility was

$$u_p(a) = \delta M_p(a) + 5I[M_{p,a} \geq 0] - 0.03(\alpha^2 + \beta^2),$$

where $\delta M_p(a)$ is the change in clean-minus-conflict candidate margin and the indicator supplies a crossing bonus. For non-closure prompts,

$$u_p(a) = -|\delta M_p(a)| - 0.10(\alpha^2 + \beta^2).$$

The policy was trained to predict the utility-maximizing action from the same 15 pre-intervention features used above. This objective searches for a bounded action that crosses the declared candidate boundary in closure while penalizing movement and dose outside closure; it does not optimize correctness.

For prompt p , let $\pi_{p,l}(k)$ be the intervention-class posterior for class k at intervention layer $l \in L_I$. Mean closure support and normalized posterior entropy were

$$\bar{\pi}_p = |L_I|^{-1} \sum_{l \in L_I} \pi_{p,l}(\text{closure}), \quad \bar{H}_p = |L_I|^{-1} \sum_{l \in L_I} \frac{-\sum_k \pi_{p,l}(k) \log \pi_{p,l}(k)}{\log 3}.$$

The gate was

$$g_p = I[\bar{\pi}_p \geq 0.50 \wedge \bar{H}_p \leq 0.90].$$

The probability and entropy thresholds were declared in the experiment protocol before held-out execution. No grid search, threshold sweep or held-out outcome tuning was performed. The reported result therefore characterizes the operating point (0.50, 0.90), not an optimized gate or a threshold-robustness claim. When $g_p = 0$, every intervention-layer action was set to (0, 0). When $g_p = 1$, the predicted model-specific action vector was executed. The primary endpoint was a strict full-vocabulary top-1 continuation change from the declared conflict token to the declared clean token. The damage guards were any full-vocabulary top-1 change among the 203 non-closure prompts and intervention-layer state-norm ratios. Clean denotes the unperturbed-chain candidate and is generally not the rule-consistent answer in closure prompts.

The confidence-gated controller crossed the declared boundary in 17/87, 13/87 and 9/87 held-out closure prompts in Qwen, Llama and Gemma. Non-closure top-1 changed in 0/203, 2/203 and 0/203 prompts, respectively. Without the final abstention gate, strict crossings were 19/87, 13/87 and 9/87, while non-closure changes were 3.45%, 1.48% and 0.49%. These results support output-boundary causal leverage at one declared operating point across the three testbeds. They do not establish answer-correctness improvement, a safety guarantee or deployment control.

To test whether the effect required precise action-to-trajectory assignment, each safety-matched repeat permuted the complete model-specific action vector without fixed points among gate-open prompts within the same closure condition. This preserved the gate-open set, all gate-closed zero actions, total action budget and every condition-by-layer (α, β) multiset. Strict crossings were defined relative to each prompt’s unperturbed top-1 output. Across 50 repeats, null crossing counts ranged from 15 to 20 in Qwen, 12 to 16 in Llama and 6 to 10 in Gemma. Their means were 17.64, 14.18 and 7.90. The one-sided empirical values were 0.804, 0.922 and 0.255, respectively.

The matched null therefore does not support fine-grained action-to-trajectory dose specificity in any checkpoint. A separately frozen component-ablation protocol retained the same gate-open sets and training fits. Operator-only actions kept each real gated α and set $\beta = 0$; precursor-only actions kept β and set $\alpha = 0$; the uniform control assigned (1.0, 1.2), the maximum existing library action, to every gate-open prompt. Negative and positive references reproduced all previous top-1 token IDs exactly before the ablations were interpreted.

Supplementary Methods Table 3 | Confidence-gated actuator component attribution.

Checkpoint	Gated policy: closure flips / non-closure changes	Operator only	Precursor only	Uniform maximum on same gate
Qwen	17/87 / 0/203	17/87 / 0/203	0/87 / 0/203	22/87 / 0/203
Llama	13/87 / 2/203	14/87 / 2/203	1/87 / 0/203	15/87 / 2/203
Gemma	9/87 / 0/203	8/87 / 0/203	0/87 / 0/203	10/87 / 1/203

Across checkpoints, the gated policy and operator-only ablation each produced 39 strict crossings and 2 non-closure changes; precursor only produced 1 and 0, and the uniform maximum produced 47 and 3. All intervened states were finite and the largest recorded state-norm ratio was below 1.10. The supported algorithmic object is therefore more specific than the original two-component description: confidence-based abstention plus the bounded local transition operator carries the principal output-boundary leverage at the declared operating point. The precursor component alone is insufficient. The Llama and uniform-Gemma damage counts show that the same gate does not guarantee zero collateral output change across checkpoints. These comparisons do not identify a universal decomposition or establish correctness control.

Executable procedure: confidence-gated local transition control

```

split complete relation graphs into training and held-out sets

fit the candidate-separation coordinate, transition operators and classifiers on training graphs

execute every bounded action on training prompts and assign the utility-maximizing action label

for each held-out prompt:

    compute layerwise mechanism posteriors from pre-intervention features

    open the gate only if mean closure support is at least 0.50 and mean entropy is at most 0.90

    if the gate is closed, apply the identity action at every intervention layer

    if the gate is open, apply the predicted bounded operator and precursor weights

    evaluate strict full-vocabulary top-1 candidate crossing and non-closure output change

compare the real assignment with safety-matched action-vector reassignments

```

13. Human-AI hypothesis-to-audit workflow

This section records research provenance and responsibility. The workflow was not evaluated as a productivity intervention: the development interval records achieved throughput in this case but does not identify the counterfactual contribution of AI assistance. The collaboration used a repeated six-step loop.

1. **Human question or contradiction:** identify a mechanistic question, evidentiary gap or failed interpretation.
2. **Executable proposal:** use AI assistance to translate the question into candidate metrics, controls, scripts or consistency checks.
3. **Local execution:** run analyses against stated checkpoints and archived data.
4. **Evidence audit:** compare outputs with leakage restrictions, nulls, source files and rival explanations.
5. **Human decision:** retain, revise or reject the object, metric, interpretation or claim.
6. **Trace-preserving iteration:** archive scripts, outputs, negative results and the reason for the next test.

AI tools assisted with project-record retrieval, code scaffolding and debugging, batch planning, data-table organization, deterministic R plotting scripts, consistency checks and language editing. The human researcher selected the scientific problem, changed the empirical object, decided which controls were decisive, interpreted failures, approved claim boundaries and remained accountable for every reported output. AI suggestions were operational drafts or candidate hypotheses, not evidence.

The primary trajectory-mapping, cross-checkpoint, geometric and intervention code sequence was developed from 2 to 6 June 2026. A proxy audit on 10 June separated direct hidden-state, vocabulary-facing and construction-aligned objects and forced correction of earlier interpretations. These dates document development order and the achieved interval, but they do not quantify productivity relative to a no-AI process, establish scientific quality or mark completion of every final run. Final archive metadata should additionally preserve execution times, hashes and environments.

14. Responsibility matrix

Function	Human authority	AI-assisted operation	Auditable artefact
Problem and object	Define question; select and revise process object	Retrieve prior discussions; compare formulations	Conversation archive and provenance ledger
Theory	Select source ideas and limits; reject misleading analogies	Organize concept mappings and draft formulas	Concept-provenance table
Experiment	Specify controls, nulls and acceptance criteria	Draft code scaffolds and parameter sweeps	Versioned scripts and configuration tables
Falsification	Interpret contradictions and decide theory change	Locate inconsistencies and propose follow-up checks	Negative-result and correction records
Data and figures	Select quantities; verify source mapping; approve claims	Assemble tables and deterministic R scripts	Source Data manifest and figure audit
Manuscript	Set evidence hierarchy, scope and final language	Terminology, formatting and language support	Versioned manuscript and audit report
Accountability	Approve and assume responsibility	No approval or authorship authority	Author Contributions and AI disclosure

15. Research independence and disclosure completeness

The study was undertaken independently and was not an internal project of any listed organization. It used no employer data, computing infrastructure, internal systems, funding, research facilities or personnel support and did not involve organizational review or approval. H.Z. personally provided or paid for all local computing, storage,

software subscriptions and other research expenses. Y.W., R.S. and W.C. contributed in their personal capacities as independent research collaborators. Affiliations identify the authors and do not imply employer sponsorship, endorsement or responsibility for the work.

The author names, affiliations, corresponding-author details, ORCID, acknowledgements, funding statement, narrative author contributions and competing-interests declaration were carried forward from the preprint record and checked against the main manuscript. The submitted scientific figures were generated from retained data and deterministic R code rather than an image-generation system.

AI product families and providers are disclosed in the main Methods. The consumer interfaces used during June and July 2026 did not consistently expose stable underlying model build identifiers, and interface-level task identifiers were not retained as scientific provenance; these fields are therefore not reconstructed. Scientific provenance instead rests on the executable scripts, fixed inputs, environment records, null specifications, retained outputs, source mappings and file-level hashes included in the archive. The public repository and Zenodo concept records are identified in the main Data and Code availability statements. Formal CRediT labels are supplied as submission metadata and do not replace the narrative author-contribution statement.

16. Additional architecture, order, metric and training controls

16.1 Random fixed-map relational continuity

Prompt states were propagated through fixed random mappings matched to the dimensional schedules of representative Qwen, Llama and Gemma configurations. The audit crossed three task families (lexical categories, addition and four-choice ARC-Challenge) with two independent seeds. The tested quantity was a cross-depth relation-preservation contrast computed from independently partitioned prompt sets and compared with lag-matched, block-donor and independent-map nulls. All 18 declared architecture-task-seed conditions passed the positive-contrast gate. The accepted claim is architecture-compatible relational continuity. Learned semantics, native-order privilege, endpoint direction and a least-action path law do not follow from this result.

16.2 Reordered execution and functional fidelity

The same trained Qwen checkpoint was executed in native, reversed and fixed-permutation block order. Geometry was compared with native full-vocabulary top-1 output, declared-candidate agreement and Jensen-Shannon divergence. Reversed and fixed-permutation executions retained positive chord contrasts but had zero full-vocabulary top-1 agreement with native execution; candidate agreement was 0.479 and mean Jensen-Shannon divergence was approximately 0.692. This falsified geometric straightness as a sufficient statistic for native computation. The retained claim is that realized order is functionally consequential even when a coarse geometric statistic remains positive.

16.3 Incremental path-history prediction

For each task-checkpoint condition, group-held-out current-state predictors were compared with otherwise matched predictors augmented by ordered-prefix summaries. Passing required a positive incremental held-out gain above the declared order-null threshold. Six of 18 initial conditions passed, all in addition; lexical prompts passed 0/9. An independently generated mixed-arithmetic replication passed 0/9. The relation-graph final-margin forecast remains valid within its declared construction, but the tested prefix summaries do not establish a task-general path-memory law. A compatible explanation is that the current high-dimensional state already compresses much of the usable history; this explanation has not yet been tested as an exact Markov property.

16.4 Radial sensitivity of the Llama statistic

Llama residual trajectories were decomposed into unit directions and state norms. Direction-only trajectories passed none of seven task gates, whereas norm-only fixed-direction trajectories passed all seven. A frozen radial exponent $\gamma \in \{0, 0.25, 0.5, 0.75, 1, 1.25\}$ produced monotonic metric changes in all seven tasks. The accepted claim is radial sensitivity under the declared metric and normalization. The statistic does not independently identify semantic direction, function or an architecture-wide law.

16.5 Pythia component compatibility

Embeddings (E), transformer blocks (B), final normalization (N) and output head (W) were swapped between initialized and trained Pythia-160m checkpoints in the complete 2^4 factorial design. All combinations were evaluated on the same 53 held-out lexical cloze prompts. Accuracy rose from 0.472 with all initialized components to 0.849 with all trained components; the best isolated trained component reached 0.604. The fully trained margin exceeded the additive component prediction by 2.621 (95% bootstrap confidence interval, 1.252-3.987). The supported claim is distributed compatibility among trained input, transition and output components in this checkpoint and task. Hybrid combinations are diagnostic interventions, not natural training states.

16.6 Training-time task coordinates

Seven Pythia checkpoints were evaluated with a model-native candidate coordinate and a coordinate fitted once at the final checkpoint. Model-native order gain was absent at initialization, appeared from step 1,000 and correlated with held-out accuracy (Spearman $\rho = 0.75$). The fixed final coordinate transferred with macro F1 0.041 at steps 0 and 128, 0.701 at step 1,000 and 1.0 by step 16,000. A coordinate refitted within each checkpoint already separated categories perfectly at initialization and therefore failed the learning-evidence gate. These results support training-associated task organization in fixed coordinates while rejecting random-initialization linear separability as evidence of learned semantics.

Supplementary Methods Table 4 | Claim-boundary audit.

Supported statement	Boundary
Ordered hidden states provide a common within-checkpoint process object	Engineering components still implement the process; cross-checkpoint coordinates are not assumed identical
Direct hidden and hidden-flow features provide stronger mechanism coordinates than readbacks in the tested audits	Readbacks remain useful diagnostics; the construction-aligned decision profile is not independent evidence
Exploratory functional windows retain prompt-condition-class information on new graph instantiations	The joint prospective recurrence gate failed: topology separation reversed in all checkpoints, downstream coupling missed its null in Llama and only Gemma retained top-quartile window location
ΔU summarizes candidate-path separation	It is construction-defined, not an energy, final-answer label or universal order parameter
O_t^{cont} improves closure	It is descriptive when adjacent target-layer information enters its construction; corrected target-excluded bilinear estimates retained 25.1% of the observed-coordinate gain and remain an exploratory predictive boundary
Hidden paths are chord-directed relative to endpoint-preserving shuffles	The result is metric-dependent and does not establish exact geodesics or least action
Chord-directed geometry recurs in lexical, addition and four-choice ARC tasks	Task-specific coordinate transfer is partial, and the numerical relation-graph ΔU axis was not transferred

Continued on next page

Supported statement	Boundary
Ordered partial histories forecast the final candidate margin in held-out relation-graph groups Layer order yields chord-directed geometry in trained and randomly initialized Qwen, Llama and Gemma architectures	Cross-task ordered-prefix summaries did not generally add prediction beyond current state; no universal path-memory law is supported Random-initialization effects were larger; geometry alone is architecture-compatible and does not identify learned inference organization
Fixed random maps preserve prompt relations across depth under three stronger nulls Reverse and permuted executions retain positive geometry but lose native outputs Pythia task function depends on trained $E/B/W$ compatibility and fixed task coordinates emerge during training	Relation preservation does not establish semantics, native-order privilege or task function Native order is functionally consequential; straightness is not sufficient for fidelity One small model, one training run and one lexical diagnostic do not establish a universal training law or complete representational geometry
Llama raw alignment is strongly radial-sensitive	Metric sensitivity does not make norm scale the mechanism or establish semantic direction
Local transition operators shift internal dominance beyond fixed and shuffled controls Held-out internal displacement predicts candidate-margin movement and pooled crossing risk Confidence-gated local transition control crosses the emitted-token candidate boundary in all three checkpoints	Internal selectivity does not by itself establish an emitted-token change Original-budget crossings were checkpoint-dependent; the map is task- and candidate-defined Strict flips were 17/87, 13/87 and 9/87; operator-only ablation preserved 17/87, 14/87 and 8/87 whereas precursor-only produced 0/87, 1/87 and 0/87; no checkpoint showed fine-grained assignment specificity; this is not answer correctness, safety or deployment control
An AI-assisted workflow completed a compressed hypothesis-to-audit sequence	This single case has no no-AI counterfactual and does not identify a causal productivity gain; scientific object choice, falsification decisions, interpretation and accountability remained human